MULTI-MODAL PSYCHOPATHOLOGY ASSESSMENT FOR STATE-OF-MIND PREDICTION

A project report submitted to the

DEPARTMENT OF COMPUTER APPLICATIONS

in partial fulfilment of the requirements for the award of the degree of

MASTER OF SCIENCE

In

APPLIED DATA SCIENCE

By
JAWAHAR S
REG. NO. RA2432014010160

Under the guidance of

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
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Kattankulathur – 603203

APRIL – 2026

BONAFIDE CERTIFICATE

This is to certify that the project report titled **"Multi-modal Psychopathology Assessment for state-of-mind Prediction**" is a Bonafide work carried out by **JAWAHAR S (RA24320140160)** under my supervision for the award of the Degree of Master of Science in Applied Data Science. To my knowledge the work reported herein is the original work done by this student.

Dr. Belina V J Sara S
Assistant Professor,
Department of Computer Applications
(GUIDE)

Dr. P J Arul Leena Rose
Professor & Head,
Department of Computer Applications

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION OF ASSOCIATION OF RESEARCH PROJECT WITH SUSTAINABLE DEVELOPMENT GOALS

This is to certify that the research project entitled "**Multi-modal Psychopathology Assessment for state-of-mind Prediction"** carried out by **Mr. JAWAHAR S** under the supervision of **Dr. Belina V J Sara S** in partial fulfilment of the requirement for the award of Post-Graduation program has been significantly or potentially associated with SDG Goal No **03 (THREE)** titled **GOOD HEALTH AND WELL- BEING.**

This study has clearly shown the extent to which its goals and objectives have been met in terms of filling the research gaps, identifying needs, resolving problems, and developing innovative solutions locally for achieving the above-mentioned SDG on a National and/or on an international level.

SIGNATURE OF THE STUDENT **GUIDE**

HEAD OF THE DEPARTMENT

ACKNOWLEDGEMENT

With profound gratitude to the ALMIGHTY, I take this chance to thank people who helped me to complete this project.

I take this as a right opportunity to say THANKS to my parents who are there to stand with me always with the words “YOU CAN”.

I wish to express my sincere gratitude to **Dr. T.R. Paarivendhar**, Chancellor, SRM Institute of Science & Technology who gave us the platform to establish me to reach greater heights.

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I convey my gratitude to all the faculty members of the department who extended their support through valuable comments and suggestions during the reviews.

A great note of gratitude to friends and people who are known and unknown to me who helped in carrying out this project work a successful one.

JAWAHAR S

PROJECT COMPLETION CERTIFICATE

INSPIRE
Softech Solutions

(A TN Govt Registered, ISO 9001-2015 Certified & AICTE Associated Organization)

CERTIFICATE OF COMPLETION

This is to certify that **JAWAHAR S** has successfully completed project titled "**Multi-modal Psychopathology Assessment for State-of-Mind Prediction**" at Inspire Softech Solutions, Chennai - 600 032.

The project was carried out from December 2025 to March 2026.

We acknowledge his efforts and dedication toward the successful completion of the project and wish him success in his future endeavours.

For Inspire Softech Solutions.,

Authorised Signatory

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Traditional psychopathology assessments rely heavily on subjective clinical interviews, which are often susceptible to patient bias and the conscious masking of true emotional states. To address the need for objective, continuous emotional monitoring, this project presents a Multi-modal Psychopathology Assessment system designed for robust state-of-mind prediction. The proposed architecture fuses visual and acoustic modalities to provide a comprehensive evaluation of a subject's affective state, mitigating the false positives inherent in single-modality systems. The visual channel employs a custom 2D Convolutional Neural Network (CNN) trained on the FER-2013 dataset to classify seven distinct facial expressions extracted from video frames. Concurrently, the auditory channel utilizes an advanced, hybrid 1D CNN-LSTM network trained on the RAVDESS dataset. Addressing early developmental limitations, the robust auditory model incorporates targeted data augmentation—such as noise injection, pitch shifting, and time stretching—and utilizes Mel-frequency cepstral coefficients (MFCCs) to capture long-term temporal and prosodic dependencies across eight acoustic emotion classes. Furthermore, the system integrates a K-Means clustering algorithm for speaker diarization, allowing the application to isolate and track emotions for specific individuals in multi-speaker environments. These dual independent streams are synchronized within a Streamlit-based web application. The integration hub utilizes a late-fusion algorithmic strategy, actively comparing synchronous visual and auditory predictions to calculate emotional congruence and flag incongruent affect (e.g., a smiling face paired with a stressed voice). The final output is rendered through an interactive, browser-based multimodal player and dynamic Plotly visualizations. Ultimately, this platform equips mental health professionals with detailed, chronological insights regarding a patient's emotional fluctuations, functioning as an impartial clinical decision tool during telehealth consultations and enabling long-term psychological, psychiatric and mental health tracking to support comprehensive patient wellness.

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